BULE HORA UNIVERSITY

COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY

INDUSTRIAL PROJECT I PROPOSAL

Title: Online Attendance Management System for Bule Hora University

Group 2 Member

No.	Name	Id no.
1.	Kakush Belete	WU/3726/13
2.	Deno Nenko	WU/3752/13
3.	Genale Nagesa	WU/3768/13
4.	Miesa Zenebe	WU/3800/13
5.	Danbi Robe	WU/3566/13
6.	Adula Boko	WU/3662/13
7.	Rahel Bayisa	WU/3744/13

Bule Hora, Ethiopia

Table of Contents

List of Tables ii

1. Introduction 1

1.1. Background 2

1.2. Statement of the Problem 2

1.3. Objectives 3

1.3.1. General Objective 3

1.3.2. Specific Objectives 3

1.4. Significance of the Study 4

1.5. Feasibility 4

1.5.1. Economic Feasibility 4

1.5.2. Technical Feasibility 4

1.5.3. Time Feasibility 5

1.6. Scope and Limitations of the Study 5

1.6.1. Scope of the Project 5

1.6.2. Limitation of the Project 5

1.7. Methodology 6

1.7.1. Requirement Gathering Methods 6

1.7.2. Requirement Modeling 6

1.7.3. Tools and Technologies 7

1.8. Work Plan 7

1.9. Project Budget 8

1.10. Conclusion 8

List of Tables

Table 1 Project Work plan 7
Table 2 Project Budget..... 8

1. Introduction

Attendance management is an important component of academic administration in higher education institutions. Monitoring student attendance helps instructors evaluate student participation, identify students who frequently miss classes, and ensure that academic regulations related to attendance are properly enforced. Attendance records also play a role in determining student eligibility for examinations and continuous assessments.

In many universities, attendance is still recorded manually using paper registers or simple spreadsheets. While these methods may be easy to implement, they often create several challenges when dealing with large numbers of students and courses. Manual attendance recording requires instructors to spend valuable classroom time marking attendance, and maintaining attendance records over long periods can become difficult.

Furthermore, manual attendance systems can result in errors, missing records, and difficulties in retrieving historical attendance data. It may also be challenging for department administrators or university management to generate attendance reports or monitor student participation across multiple courses.

With the advancement of web-based information systems, universities can improve administrative efficiency by automating attendance management. An Online Attendance Management System provides a digital platform where instructors can record attendance electronically and store records in a centralized database. The system can also generate attendance reports, track student attendance patterns, and allow administrators to monitor attendance information more effectively.

Therefore, the development of an Online Attendance Management System for Bule Hora University will improve the efficiency, accuracy, and accessibility of attendance records while reducing manual work and administrative complexity.

1.1. Background

Higher education institutions are responsible for maintaining accurate academic records for students. Among these records, attendance information is essential because it reflects student participation in lectures, laboratory sessions, and other academic activities.

In many universities, attendance is recorded using traditional methods such as paper registers, printed attendance sheets, or manual entry into spreadsheets. While these methods have been used for many years, they become inefficient when the number of students increases and when multiple courses must be managed simultaneously.

Manual attendance systems have several disadvantages. For example, attendance sheets may be lost or damaged, and manually recorded data may contain errors. Additionally, generating attendance reports for individual students or entire classes requires significant time and effort.

Modern information technology solutions offer opportunities to improve the way attendance is managed in educational institutions. Web-based systems allow data to be stored securely in centralized databases and accessed easily by authorized users.

An Online Attendance Management System enables instructors to record attendance using a computer or mobile device. The system automatically stores attendance records and can generate reports that show attendance percentages, absentee lists, and attendance trends. By adopting such a system, Bule Hora University can enhance the management of attendance records and support better academic monitoring.

1.2. Statement of the Problem

The current attendance management practices used in many educational institutions face several challenges. Manual attendance recording requires instructors to mark attendance for each student during class time, which can be time consuming and reduce valuable teaching time.

Another problem associated with manual attendance systems is the possibility of human errors. Instructors may accidentally mark incorrect attendance, skip students, or record

incomplete information. Over time, these errors can affect the accuracy of attendance reports.

In addition, manual attendance records are often stored in paper documents or separate files, which makes it difficult to retrieve historical data when needed. Administrative staff may also face difficulties when generating reports about student attendance performance.

Moreover, monitoring attendance across multiple courses and departments becomes challenging when data is not stored in a centralized system. This lack of integration can affect the ability of university administrators to identify students with poor attendance or enforce attendance regulations effectively.

These challenges highlight the need for a modern online system that can automate the recording, storage, and management of attendance records. An Online Attendance Management System can address these problems by providing an efficient, reliable, and accessible platform for managing attendance information.

1.3. Objectives

1.3.1. General Objective

To design and develop an Online Attendance Management System for Bule Hora University that automates the process of recording, storing, and monitoring student attendance.

1.3.2. Specific Objectives

- To study the existing attendance recording system used in the university.
- To identify the limitations and challenges associated with manual attendance management.
- To design a web-based attendance management system that supports efficient attendance recording.
- To develop a centralized database for storing attendance records.
- To implement modules for student registration, course management, and attendance tracking.
- To generate reports that show attendance statistics for students and courses.

- To improve the efficiency and accuracy of attendance management in the university.

1.4. Significance of the Study

The proposed system will provide several benefits to the university community.

First, it will improve the efficiency of attendance recording, allowing instructors to record attendance quickly using a digital interface. Second, it will reduce errors associated with manual attendance records.

The system will also make it easier for administrators to generate attendance reports and monitor student participation. In addition, the centralized database will ensure that attendance records are securely stored and easily accessible. Overall, the system will support the digital transformation of academic administration at Bule Hora University.

1.5. Feasibility

A feasibility study is conducted to determine whether the proposed system is practical and achievable. It considers economic, technical, and time feasibility.

1.5.1. Economic Feasibility

The proposed system is economically feasible because it relies on open-source software technologies that do not require licensing fees. Development tools such as HTML, CSS, JavaScript, PHP, and MySQL are freely available.

In addition, the university can use its existing computer infrastructure to deploy the system, which reduces the need for additional hardware investment. Therefore, the overall cost of developing and implementing the system is minimal.

1.5.2. Technical Feasibility

The project is technically feasible because the required technologies and tools are widely available and commonly used in web application development. The development team possesses the necessary technical knowledge and programming skills to implement the system successfully.

Furthermore, the university already has basic computer infrastructure and internet connectivity that can support the deployment of the system.

1.5.3. Time Feasibility

The project is feasible in terms of time because it can be completed within the academic project period. The development process will follow structured phases including requirement analysis, system design, coding, testing, and documentation.

Proper planning and task allocation will ensure that each phase is completed within the allocated time. With proper planning and development effort, the project can be completed within the allocated academic project duration, which is usually 5–6 months or two semesters.

1.6. Scope and Limitations of the Study

1.6.1. Scope of the Project

The Online Attendance Management System will focus on managing attendance records for students at Bule Hora University.

The system will include the following features:

- Student registration and management
- Instructor account management
- Course and class management
- Recording daily attendance
- Viewing and updating attendance records
- Generating attendance reports
- Monitoring student attendance statistics

1.6.2. Limitation of the Project

Although the system will improve attendance management, some limitations may exist:

- The system will be developed specifically for Bule Hora University.
- Biometric or facial recognition attendance features may not be included in the initial version.
- The system will require internet access to function effectively.

1.7. Methodology

1.7.1. Requirement Gathering Methods

To understand the requirements of the proposed system, different data collection methods will be used.

- **Interview**

Interviews will be conducted with instructors, department heads, and administrative staff who are responsible for managing student attendance. These interviews will help gather information about the current system and identify its strengths and weaknesses.

- **Observation**

Observation will be used to study how attendance is currently recorded during classroom sessions. This method will help identify inefficiencies and opportunities for improvement.

- **Document Analysis**

Existing attendance records, forms, and related documents will be reviewed to determine the type of data that needs to be stored in the proposed system.

1.7.2. Requirement Modeling

The proposed system will use the Object-Oriented modeling approach for system analysis and design. Object-oriented modeling represents the system as a collection of interacting objects, each with its own attributes and behaviors.

Unified Modeling Language diagrams will be used to represent system structure and interactions. These diagrams may include:

- Use Case Diagram
- Activity Diagram
- Sequence Diagram
- Class Diagram

This approach provides a clear visualization of system functionality and helps developers understand system requirements.

1.7.3. Tools and Technologies

The following technologies will be used to develop the system:

- ✓ **Frontend:** HTML, CSS, JavaScript
- ✓ **Backend:** PHP
- ✓ **Database:** MySQL
- ✓ **Server Environment:** XAMPP
- ✓ **Development Tools:** Notepad++

1.8. Work Plan

Table 1 Project Work plan

Activity	Jan	Feb	Mar	Apr	Ma	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Project Topic Selection												
Proposal Preparation												
Requirement Analysis												
System Design												
System Development												
System Testing												
Documentation Writing												
Final Report Preparation												
Presentation Preparation												
Final Project Submission												

1.9. Project Budget

Table 2 Project Budget

No	Item Description	Quantity	Unit Cost (ETB)	Total Cost (ETB)
1	Internet Access	3 Months	1600	4800
2	Printing and Photocopy	210 Pages	10	2100
3	Flash Disk / Storage Device	1	1000	1000
4	Stationery (Pen, Notebook, etc.)	1 Set	550	550
6	Transportation	5 Trips	110	550
	Total Budget			70**00**

1.10. Conclusion

The Online Attendance Management System will provide an effective solution for managing attendance records at Bule Hora University. By automating attendance recording and storing data in a centralized database, the system will improve efficiency, reduce errors, and simplify the generation of attendance reports.

The implementation of this system will support better monitoring of student participation and enhance the overall management of academic activities within the university.